Einstein's life as a professor was not easy. He found it irritating to have to give regular lectures on topics not close to his research. He could summon neither the energy to prepare such lectures well nor the enthusiasm to make them scintillate, even though when lecturing on topics dear to his heart, he was brilliant. Einstein was now a full-fledged member of Europe's academic circle, but he was paying a price. Despite this price, his research in the realm of the small moved forward impressively, producing insights that later would win him the Nobel Prize (see Box 4.1).

Then, in mid-1911, Einstein's fascination with the small waned and his attention returned to gravity, with which he would struggle almost full time until his triumphant formulation of general relativity in November 1915.

The initial focus of Einstein's gravitational struggle was *tidal gravitational forces.*

Tidal Gravity and Spacetime Curvature

Imagine yourself an astronaut out in space, far above the Earth's equator, and falling freely toward it. Although, as you fall, you will not feel your own weight, you will, in fact, feel some tiny, residual effects of gravity. Those residuals are called "tidal gravity," and they can be understood by thinking about the gravitational forces you feel, first from the viewpoint of someone watching you from the Earth below, and then from your own viewpoint.